Staff answers for the Japan hazard case

Tuesday 9 March 2027 | P2 | 40 minutes | Year A

Humanities | LAUNCH | Spring 2 Week 3 | HUM-LAUNCH-S2-W03

Case and effects

Use USGS magnitude 9.1, depth 29 km and origin 05:46:24 UTC for the same 11 March 2011 Tohoku Japan case. A primary chain is ground shaking directly damaging buildings. A secondary chain is sea floor displacement, displaced water, tsunami and coastal flooding. Roads, rail, airports and utilities can involve both earthquake and tsunami damage; a supported mixed classification is valid. People and places also overlap.

Management and limits

Protection: resistant buildings, retrofits and coastal barriers reduce some physical risks. In 2011 some coastal defences were overtopped or damaged. Preparation: practised plans, clear routes, education and supplies support response. Early warning communicates detected danger and can support prepared action. No measure guarantees safety; the strength of a combined approach is an evidence-based judgement, not a claim that warnings predict exact earthquakes.

Timing precision

Japan's earthquake early warning was issued 8.6 seconds after first P-wave detection at the nearest station. Do not state 8.6 seconds after earthquake origin. The first tsunami warning was at 05:49 UTC, about three minutes after the 05:46 earthquake. Initial tsunami estimates were too low and updated. Avoid calculating an exact delay from minute-rounded warning data.

Route judgements

Every route explains one primary and one secondary effect, one protection and one preparation measure, and a warning or prediction limit. Supported uses D1 and frames with actual help recorded. Independent links each measure to risk and explains combined use. Aspire uses the warning timeline and mixed-effect evidence to judge limits. For classroom time sorting, short term is days or weeks and long term months or years; real effects can span both.

Sensitive response

Use factual, non-graphic summaries. No casualty numbers, nuclear detail, personal testimony, disaster simulation or family prompts. Use the agreed supervised pause and private re-entry. Follow the normal safeguarding route if a concern is disclosed.

Hinge: Which claim is accurate? A: We can predict an earthquake's exact date. B: Early warning detects a quake already under way; preparation helps people act. C: A sea wall stops every tsunami. Answer B.

Parallel check: A barrier is overtopped. Does this mean evacuation plans are unnecessary? Answer: no. It shows that protection has limits, so preparation and practised routes remain important. Ask each learner to explain the because link.

Offer to the Communication portfolio. Keep the learner's own idea, an actual adult audience reply, one NS and the learner's response in the same colour, on the same page or usual mode before R. E follows actual evidence banking. Record actual help. A partner's answer cannot establish both learners' understanding.

Actual observation _____ Help _____
Gap and planned follow-up _____ Actual banking _____